A Novel Wave Height Sensor

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ABSTRACT

A novel transducer for the measurement of water level is described. This transducer consists of a tantalum wire anodized in a weak citric acid solution. The anodization forms a uniform, thin layer of tantalum oxide on the surface of the wire. When partially submerged through an air–water interface, the capacitance of the wire is directly proportional to the water level. This transducer has been applied in oceanographic research buoys to measure gravity and capillary wave height. In this application, the transducer offers greater sensitivity, linearity, and ruggedness than conventional capacitive wave sensors.

1. Introduction

A variety of transducers has been developed within the oceanographic community for the measurement of ocean waves. The most common type employs a thin wire protruding through the interface. Wave height can be sensed from this wire by measuring either its resistance or its capacitance relative to the water. Resistive wire gauges are commonly used but suffer from problems of marine fouling and show an undesirable sensitivity to salinity. Resistive wave gauges operated in close proximity to each other also suffer from cross-talk problems.

Capacitive wave gauges depend on the wire and a thin dielectric isolating them from the seawater. Capacitive gauges are generally less sensitive to fouling and salinity changes. Furthermore, cross-talk effects can be minimized through the use of sensing circuits with synchronous detectors operating at different frequencies. Capacitive gauges do suffer from several problems though. Variations in the meniscus that may induce nonlinearities in transient response (Keller and Chapman 1980) while limiting frequency response (White and Tallmadge 1965; Sturm and Sorrell 1973) are a problem common to all wetted gauges. Problems specific to capacitive gauges include nonlinearities in transducer response due to nonuniformities in the dielectric. At the same time, most capacitive sensors are high-impedance devices, making them sensitive to stray capacitance. In some cases, this can complicate the absolute calibration of capacitive sensors.

In this paper we describe a capacitive wave gauge transducer consisting of a tantalum wire coated with a uniform thin layer of tantalum oxide. Tantalum oxide is an insulator and excellent dielectric material. Such a transducer, easily made through an anodization process, offers sensitivities of several microfarads per meter and linearities approaching one part per thousand. The transducers are rugged and when used in saltwater exhibit a self-healing capability that automatically repairs any damage that may occur to the tantalum oxide dielectric.

Section 2 provides an overview of the entire wave gauge system, including the sensor and associated electronics. Section 3 describes the anodization process for creating the tantalum wire transducers. This is described in sufficient detail to enable interested investigators to construct their own wave gauge sensors. In section 4 we describe the wave gauge electronics design that we are currently using. Finally, the procedures for calibration and some example data are discussed in section 5.

2. Tantalum wave gauge system overview

The Applied Physics Laboratory (APL) wave gauge system consists of a precision tantalum capacitive wave height sensor and a wave gauge electronics package. The tantalum wire sensor has tantalum oxide coating (Ta_2O_5) that is grown in the laboratory prior to deployment by anodizing clean tantalum wire in a weak citric acid solution. While tantalum is a metal, Ta_2O_5 is a good insulator with a high dielectric constant. A very thin and uniform coating can be easily made providing for a highly sensitive but extremely linear sensor. Furthermore, laboratory investigations have shown that meniscus effects on these coatings are relatively small (Keller and Chapman 1980).

The capacitance of this sensor is detected using a simple, low-power circuit made of commonly available

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components. The circuit uses a square wave oscillator to periodically charge and discharge the capacitive sensor. The time it takes to discharge the sensor to a threshold voltage determines the width of the output pulse. This pulse width, which is proportional to the sensor capacitance, can either be detected directly using a digital interface or it can be filtered to provide an analog output. The circuit maintains a mean dc potential of several volts on the wire. This voltage acts to re-anodize and heal the dielectric should it be scratched or damaged during the deployment. This self-healing mechanism allows the extremely thin Ta_2O_5 coatings to be used with a high degree of success.

The wave gauge electronics package is housed in a sealed 3-cm-diameter polyvinyl chloride (PVC) tube with an underwater connector on one end and a sensor wire connector on the other. The package weighs just a few hundred grams and provides an output voltage of 0.5–4.5 V corresponding to excursions from 0 to 10 μF input capacitance. The unit draws approximately 30 mA at 12 V and will operate over a range from 8 to 15 V.

These wave gauges have been deployed on a number of platforms, including an APL-designed spar buoy. In the current spar configuration, the output of the wave gauge system is connected to a single board computer in the spar telemetry box that digitizes the analog wave gauge signal and inserts the data into the telemetry stream transmitted back to a nearby research vessel via a radio-frequency modem.

3. Anodization of the wave gauge wire

A series of experiments were conducted to determine the controlling factors in the anodization of the wave gauge wire sensors. In these experiments we studied the effects of various electrolytes, formation voltages, and initial formation current densities. The effects of these variables on sensor linearity and stability in both fresh- and saltwater were examined. While we found that the linearity and stability of these sensors are quite insensitive to the details of the formation process, we have developed a standard formula for anodizing sensor wires. This standard formula is described in this section.

The wave sensors are formed from unenamaled capacitive-grade tantalum wire (NRC Metals, Newton, MA, type TPX). The use of alternative grades of wire has not been investigated. We use 0.635-mm-diameter wire as a compromise between thickness (thinner wires have less influence on waves and exhibit higher frequency response) and strength. This grade of 0.635-mm-diamater wire has a tensile strength of about 70 kg mm^{-2} for a 4% elongation. This corresponds to a 22-kg load, more than enough load to maintain the tautness of the sensor.

The wires are cut into 5-m sections and thoroughly cleaned with a commercial trichloroethane solution (Swish Aersol Electrokleen). This cleansing removes grease and dirt left on the wire from the manufacturing process. The wires are then rinsed well in tap water and wiped dry. Care is taken not to touch the cleaned wires as oils from the skin will affect the anodization process.

The wires are then handled by the ends using surgical gloves and placed in a 4-m-long anodization bath. The bath consists of a 15-cm-diameter PVC tube that has been cut in half lengthwise to form an open trough (Fig. 1). The ends are sealed with Plexiglas semicircles that are glued to the PVC tube. A terminal strip is mounted on the outside of each end piece. A pair of semicircular Plexiglas wire supports are glued to the inside of the tube about 7 cm from each end. The supports have a series of guide holes drilled in them. At each end of the bath the wires are strung through these holes, up over the end cap, and terminated at the screw terminal strip. This scheme keeps the wires submersed and out of contact with any other surfaces. The electrical circuit within the bath is completed through a 6-mm stainless steel rod that extends the length of the bath and rests on the bottom. The connection to this rod is made with all stainless steel hardware so that only stainless steel and tantalum metals are actually in contact with the electrolyte. The presence of dissimilar metals in the bath has a detrimental effect on the anodization process.

The electrolyte is a 0.01% citric acid solution consisting of 9.8 g of anhydrous citric acid dissolved in 98 L of tap water. The electrolyte was mixed with a motor-driven Plexiglas stirrer for several hours and then left for 24 h to equilibrate to room temperature.

The current apparatus can anodize up to 11 wires at one time. All of the wires are shorted together at both ends. The anodizing current is then applied between the tantalum wires (+) and the stainless steel ground rod (−).

A constant-current or constant-voltage power supply is used to provide the anodizing current. The power supply is set to deliver a constant current equivalent to 0.5 mA cm^{-2} up to a formation voltage of 80 V. At 80 V the power supply automatically switches to a

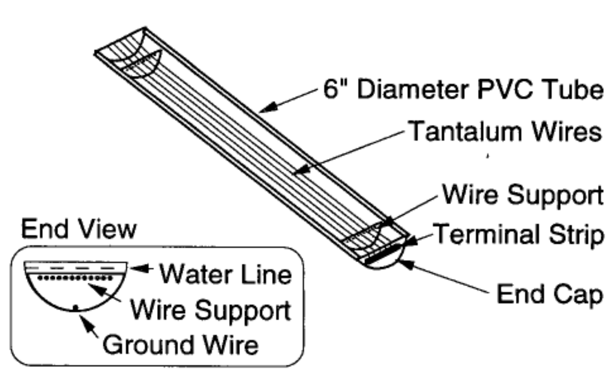

FIG. 1. Schematic of the anodizing bath.

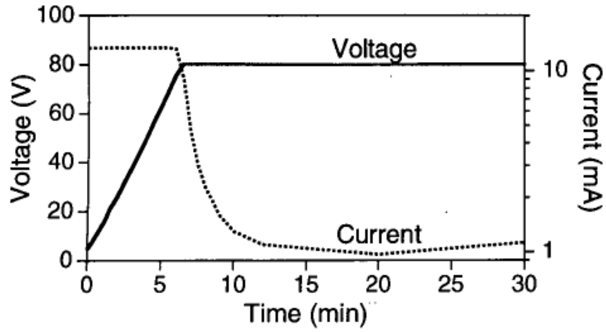

FIG. 2. Typical voltage–current characteristics of anodization process.

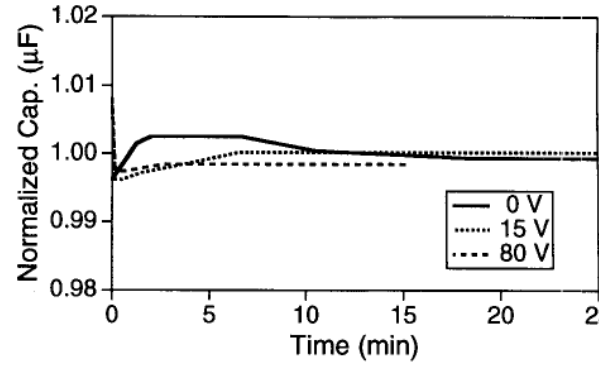

FIG. 3. Short-term aging of citric acid–anodized wires in saltwater at three different applied voltages.

constant-voltage mode and the delivered current typically falls as the anodization process continues. Figure 2 illustrates the voltage and current curves from a single anodization run.

In this particular example, 132 cm of wire having a total surface area of 26.3 cm^2 was anodized. The initial current was thus set to 13 mA. To first order, the initial growth of the oxide layer causes the voltage to rise linearly with time. When the formation voltage is reached, the power supply switches to constant voltage mode and the current quickly drops off. The current then reaches some minimum value and may actually begin to slowly increase with time.

The current does not decrease to zero due to continued migration of ions through defects in the Ta_2O_5 layer. The increase in current at later times is caused by an increase in the number of defect sites in the oxide layer due to crystallization of the Ta_2O_5 (Young 1969). The minimum current level is thus an indication of the quality of the oxide layer. The example shown is not our best example due to some minor contamination of the electrolyte solution. A simple cleaning and refilling of the bath with new electrolyte lowered the minimum current levels by an order of magnitude from those in Fig. 2.

We typically terminate the anodization process after the currents have fallen below 0.005 mA cm^{-2} or if the leakage current begins to rise. The wires are removed from the bath and carefully rinsed in freshwater to remove any remnant electrolyte. The ends of the wires are removed; only the section inside of the wire support is actually used for the wave sensor. After drying, the wires have a uniform metallic purple color. This color is a result of interference at the front and back of the oxide layer, and hence, the hue of the wires indicates the thickness of the oxide layer. Thus, uniformity of color can be used as a gross measure of oxide-layer uniformity.

Prior to calibration or use, the wires should be soaked in a saltwater bath for at least 30 min. We have found that there is some small initial drift in the capacitance of the wire when first placed in saltwater. This drift is thought to be associated with some self-limiting chemical reaction at the surface. Figure 3 shows the results of a separate test performed to examine this drift in more detail. In this test, three wires were anodized in citric acid and then placed into a saltwater bath. Two of the wires had healing voltages of 15 and 80 V applied to them. The capacitance of the wires was then measured as a function of time. It is apparent that the drift is small and the majority of drift occurs in the first 30 min, so we concluded that wires should be soaked in saltwater prior to all calibrations.

Wires created using this procedure have a capacitance of about 3.2 $\mu F\ m^{-1}$. The capacitance per unit length of the sensor is determined by the oxide thicknesses and the dielectric constant of Ta_2O_5. The oxide thickness can be controlled by adjusting the formation voltage. For dilute acid solutions, Young (1961) provides a graph showing the relationship between initial formation current density, formation voltage, and capacitance per unit area. This relationship [$VCA^{-1} = 13.1 + 0.8 \log (IA^{-1})$ in the units shown in Fig. 4] is in substantial agreement with our results as the voltage–capacitance product for our wires is 80 V $\times$ 3.2 $\mu F/20\ cm^2 = 12.8\ V\ \mu F\ cm^{-2}$. Larger values of capacitance per unit length can be obtained by applying

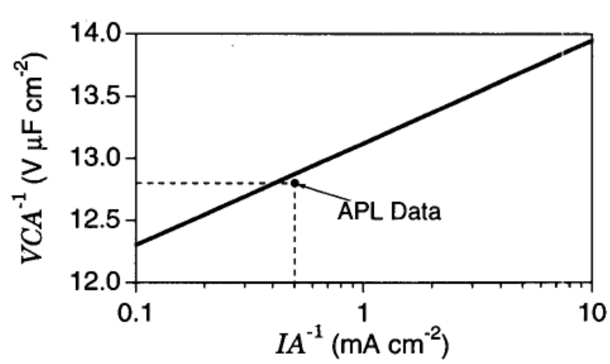

FIG. 4. Relationship between capacitance per unit area (CA^{-1}), formation voltage (V), and formation current density (IA^{-1}) for Ta_2O_5 films formed in dilute acid from Young (1961).

a smaller formation voltage, although higher leakage currents can be expected from the thinner oxide layer. In the other direction, there is a practical limit on decreasing the capacitance per unit length as the anodization process breaks down at voltages exceeding 200 V. The field strength associated with higher voltages causes increased crystallization and the eventual breakdown of the film.

Tantalum oxide films formed in this manner are in general quite robust and strongly adhere to the base metal. This can be readily demonstrated by bending an anodized wire. From the color changes of the oxide layer along the outside diameter of the bend, it is evident that the layer deforms plastically, becoming thinner where the wire is stretched and thicker where it is compressed. This leads to the restriction that the wires should not be bent before or during use, although they can be loosely rolled for transport. Wires that have been bent may exhibit strong nonlinear sensitivities in the region of the bend. We simply discard any wires that become bent prior to deployment.

4. Wave gauge sensing electronics

The capacitance of the wave gauge sensor is detected using a simple circuit that alternatively charges and discharges the sensor. The operation of this circuit, shown in Fig. 5, is easy to understand. Following the signal path starting at the upper left, the circuit begins with a simple 200-Hz RC oscillator built around a single inverter from a 74HC14 Schmitt trigger-integrated circuit. The output of this oscillator is then buffered by four other sections of the same integrated circuit (waveform A) to provide sufficient drive current to charge the sensor quickly. The buffered square wave alternatively charges the sensor through the 1N914 diode and discharges the sensor through the fixed and variable resistors. The charging occurs quickly due to the low impedance of the diode and output stages of the 74HC14; the discharge occurs at variable rates determined by the sensor capacitance and the resistance in the circuit (waveform B). The voltage across the sensor is sensed by the last section of the 74HC14, creating a variable-width pulse depending on the capacitance of the sensor (waveform C).

The remainder of the circuitry in Fig. 5 converts this pulse-width modulated signal into a dc voltage by applying the signal to a third-order low-pass filter designed about the first CA3160 operational amplifier. This filtered signal is then buffered through a second op amp that also provides offset and gain controls. Alternatively, the signal at point *C* could be sent directly to a microprocessor for a digital measurement of the pulse width.

In this example, the cutoff frequency of the filter has been set to 12 Hz. This frequency was chosen to prevent aliasing of signal or noise by the digitizer, which operates at 40 Hz. Digital filtering is then applied to the 40-Hz data prior to decimation to 20 Hz, the rate at which the data are actually recorded.

The wave gauge electronics package is housed in a sealed 3-cm-diameter PVC tube with an underwater connector for power and signal at one end and a sensor wire connector at the other. The entire unit weighs less than 500 g and draws approximately 30 mA at 12 V. The unit will operate over a range of input voltages from 8 to 15 V providing an output voltage of 0.5–4.5 V corresponding to excursions in input capacitance from 0 to 10 μF with less than 0.2% error. The circuits have proven to be stable over periods of several months. Tests indicate that the gain and offset of this circuit change by only a few percent over a wide range of temperatures.

5. System calibration

After anodizing a set of wires, one end of each wire is terminated into a fiberglass end piece (Fig. 6). This serves to seal the submerged end of the sensor wire, provides a simple and secure mechanical connection, and uniquely identifies the wires. (Each plastic end piece has a unique serial number engraved into its outer surface.) This latter feature is particularly important as it allows individual wires to be identified for calibration.

The finished sensors are calibrated in a 4-m-high wave wire calibration tank. The tank is connected to a holding tank via a pumping system that allows the water level to be raised or lowered. The water level in the tank can be measured directly through a sight tube and ruler that extends from the bottom to the top of the tank. The set of wires are held in place in the tank by a fixture consisting of a 3-m pipe with end plates for affixing the wires. Measurements are made by raising or lowering the water level to specified depths and then making measurements of the capacitance of each wire in turn. Before actual calibration, the wires are fully immersed in the saltwater of the calibration tank for at least 30 min to eliminate the short-term drift that is typically experienced at first immersion.

A typical calibration curve is shown in Fig. 7. The typical deviations of the measured capacitance are on the order of ± 0.05 μF. It is estimated that the errors associated with this calibration are 0.015 μF in capacitance and 0.5 cm in height. The height error corresponds to a capacitance error of 0.02 μF. The residual error of ± 0.04 μF is likely due to nonlinearities in the oxide thickness. This represents an error of 0.4% of full scale, a figure that is likely to be insignificant compared to the statistical fluctuations associated with sampling surface gravity waves.

It should be noted, however, that tantalum wire transducers require a positive bias voltage in normal operation, as they have nonlinear voltage-current characteristics. Tantalum is a valve metal, so called

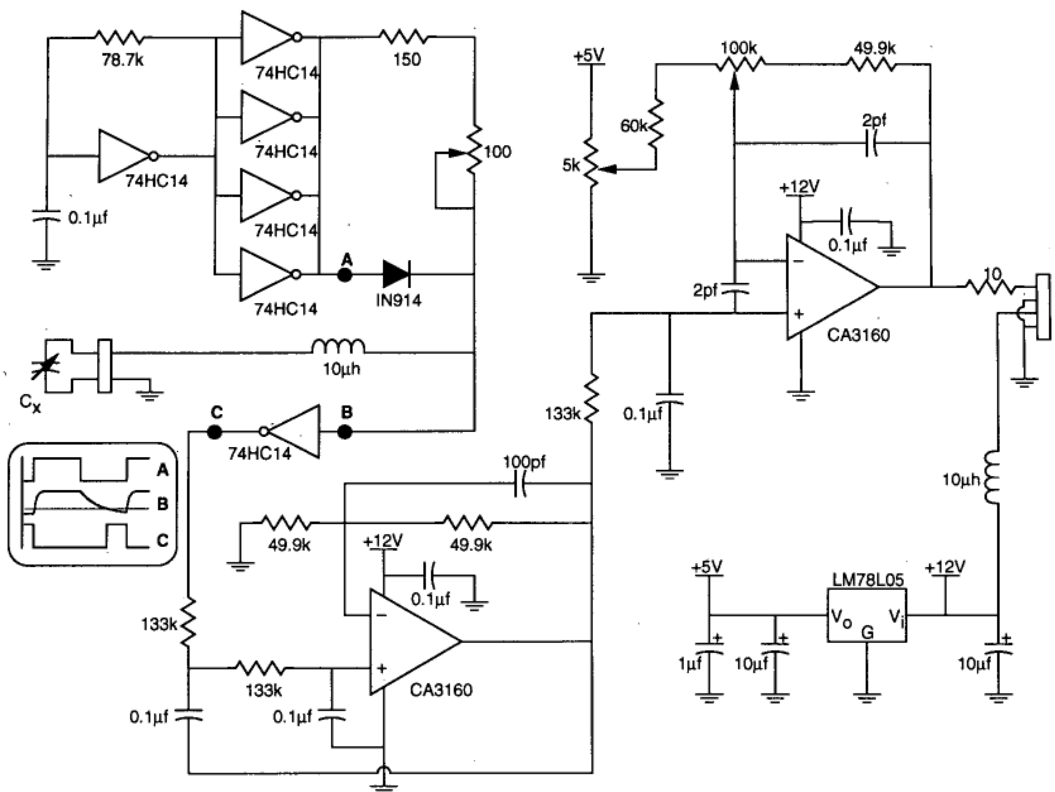

FIG. 5. Wave gauge electronics schematic diagram.

because its oxide is a rectifier when placed in an electrolyte. Current flows freely at negative voltages through the production of hydrogen in the electrolyte. At large positive voltages an ionic current begins that is responsible for the growth of an oxide layer on the metal (anodization). No current flows and the oxide layer essentially acts as an insulator for small positive voltages. For practical systems, this means that tantalum transducers should be biased so that they are always positive with respect to ground. In our system, biases as low as 5 V are used to provide for satisfactory self-healing in the case of damage.

6. Comparison with other sensors

The new wave gauge sensor we have described here has a number of demonstrable advantages over existing sensors. In the discussion below, we list these advantages in comparison with the majority of sensor types in use today: resistive wires, bare-wire impedance probes, PVC-coated wires, Teflon-coated wires, and varnish-coated magnet wire.

1) Salinity sensitivity. Resistive wires are sensitive to salinity changes that can occur in coastal environments. Tests with our capacitive sensor show no measurable sensitivity to salinity changes over a range of conductivities spanning those encountered in the ocean.

2) Cross talk. Resistive wires can suffer from cross talk when used in close proximity, as in a wave gauge

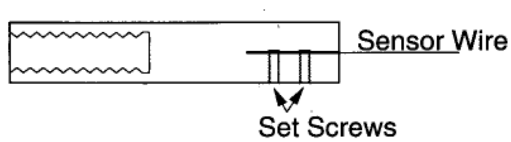

FIG. 6. Sensor wire end piece.

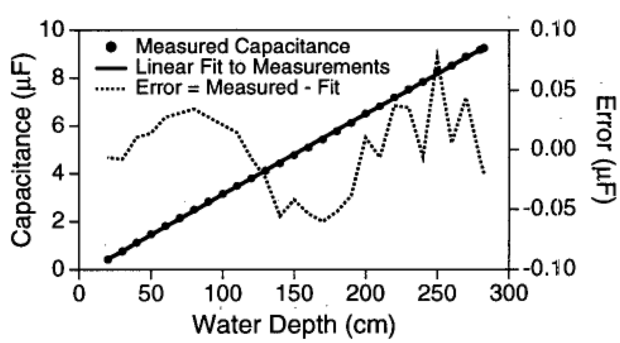

FIG. 7. Typical calibration results for a Ta_2O_5 wave sensor.

array. Bare-wire impedance probes (McGoldrick 1971), often used in tank environments, can also suffer from this problem. Tests with our capacitive system show no measurable cross talk with sensor spacings down to 2 cm. Cross talk that may occur at smaller spacings can be eliminated through time-multiplexed or detuned synchronous detection schemes.

3) Frequency response. The measurements of Keller and Chapman (1980) indicate that the frequency response of Ta_2O_5 wires is as good or better than Teflon-coated wires, both of which are clearly superior to PVC-coated wires. Frequency response is in large part driven by the size of the meniscus (Sturm and Sorrell 1973), which is governed by the surface tension of the fluid, the diameter of the sensor, and the wetting characteristics of the sensor coating. Spectral data suggest that the frequency response of the wires we use (0.635-mm diameter) is down 3 dB at about 16 Hz, which is in good agreement with the predictions of Sturm and Sorrell. We have not performed testing of response as a function of wire diameter but we expect the response to improve as the wire thickness decreases.

4) Linearity. Linearity of capacitive sensors is determined by the uniformity of the dielectric coating. Varnish-coated magnet wires typically exhibit nonlinear errors of up to 3% of full scale. This is significantly higher than the 0.4% exhibited by the Ta_2O_5 sensors. PVC and Teflon-coated wires can approach but do not exceed the linearity of the Ta_2O_5 sensors.

5) Sensitivity. PVC and Teflon-coated wires typically exhibit sensitivities of 1–10 pF cm^{-1}. The Ta_2O_5 sensor that we use has a sensitivity of 30 nF cm^{-1}, a factor of 3000 greater than PVC or Teflon. Sensitivity by itself is no advantage, but greater sensitivity does make the system less liable to effects of stray capacitance. Changes in cabling and electronics that come with age, temperature variations, and exposure to the elements can easily cause capacitance variations of a few picofarads, variations that could shift the measured level of a Teflon-coated sensor by as much as a few centimeters. In addition, the lower overall impedance of the Ta_2O_5 sensor makes it less susceptible to noise induced from nearby electromagnetic fields, a real concern for at-sea systems on board ships or buoys with RF telemetry.

6) Durability. PVC and Teflon-coated wires exhibit excellent durability of the dielectric. The Ta_2O_5 dielectric is more rugged than varnish but can be damaged by sharp metallic objects. This decreased durability is offset by the self-healing capability of the Ta_2O_5 sensor. In several dozen weeks of exposure during more than eight experiments we have never experienced a dielectric failure on a deployed tantalum wire sensor. The other factor in durability is wire strength. Common magnet wire, made from copper, is quite weak. Users report that wire breakage is a common problem in open ocean deployments. Stainless steel and tantalum wires are both significantly stronger with yield strengths of four times that of copper (Tapley 1990).

7) Repeatability. We have twice performed calibrations before and after a 2-week-long deployment. These pre- and post-test calibrations show that the calibrations of the Ta_2O_5 sensors change less than 3%. While we have no side-by-side comparisons with other sensors, this degree of repeatability is certainly good enough for all short-term oceanographic applications. The long-term repeatability of these sensors has not been tested.

Finally, the utility of the system can be illustrated with some sample data from a 1992 experiment in the New York Bight. During this 2-week experiment, three gauges were mounted on a 12-m-long spar buoy along with other instrumentation. These wave data were then split into 2-min segments for spectral analysis. An ensemble average of 10 such spectra is shown in Fig. 8.

The secondary peak at 0.06 Hz is associated with the buoyancy period of the spar buoy, which has been measured to be 16 s. At frequencies higher than this the buoy is quite stable, providing a good platform from which to make wave measurements. The noise floor due to electronics and digitization is not evident in this graph. The reduction in the slope of the frequency spectrum at frequencies of 6–10 Hz is due to the Doppler shifting of the shorter waves caused by their advection by the orbital velocities of the longer waves (Banner 1991).

7. Summary

Tantalum oxide wave gauge sensors offer several advantages including high sensitivity, stability, low sensitivity to salinity variations, and good high-frequency

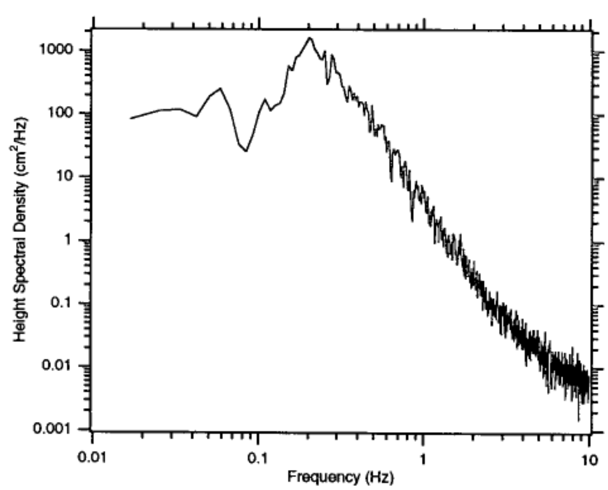

FIG. 8. Example wave height spectral density derived from the tantalum wire wave gauge system. These data were acquired from a gauge mounted on a spar buoy deployed in the New York Bight. This spectrum is an ensemble average of 10 estimates, each made from a separate 2-min data segment.

response. These sensors are easily created in the laboratory using simple apparatus. Various aspects of creating and using these sensors have been described. During several years of use in a variety of systems, these gauges have proven to be a reliable, accurate, and inexpensive means of measuring wave height.

Acknowledgments. The development of the APL wave gauge system involved the efforts of a number of people over a period of many years. The use of tantalum wire sensors (albeit anodized in saltwater) originated with Dr. Blyth Hughes' group at the Defence Research Establishment Pacific in Victoria, Canada (Hughes and Grant 1975). The use of tantalum wires was introduced to APL by Mr. Christian Keller who did much of the early engineering and design of our wave gauge systems. Keller introduced the idea of citric acid anodization and developed the APL wave gauge calibration system. Mr. Robert Miller developed the current version of the wave gauge electronics package based on an initial design of the author. The apparatus used in these experiments was constructed by the late, and sorely missed, Mr. James Allison.

We also acknowledge the help that we received from Mr. Haig Vartanian of NRC Metals. Mr. Vartanian spent considerable time on the phone explaining various aspects of metallurgy, in general, and tantalum anodization, in particular. His assistance was greatly appreciated.

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